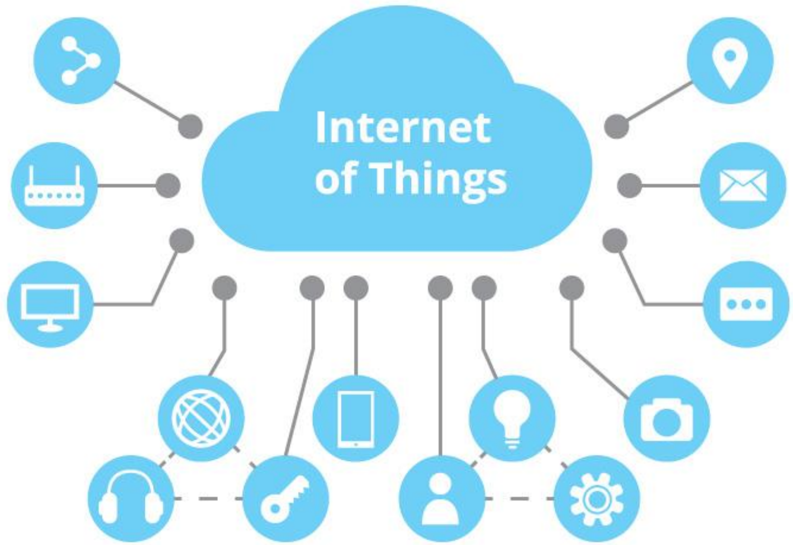
University of Economics Ho Chi Minh City
Institute of Intelligent & Interactive Technologies

Internet of Things (IoT)



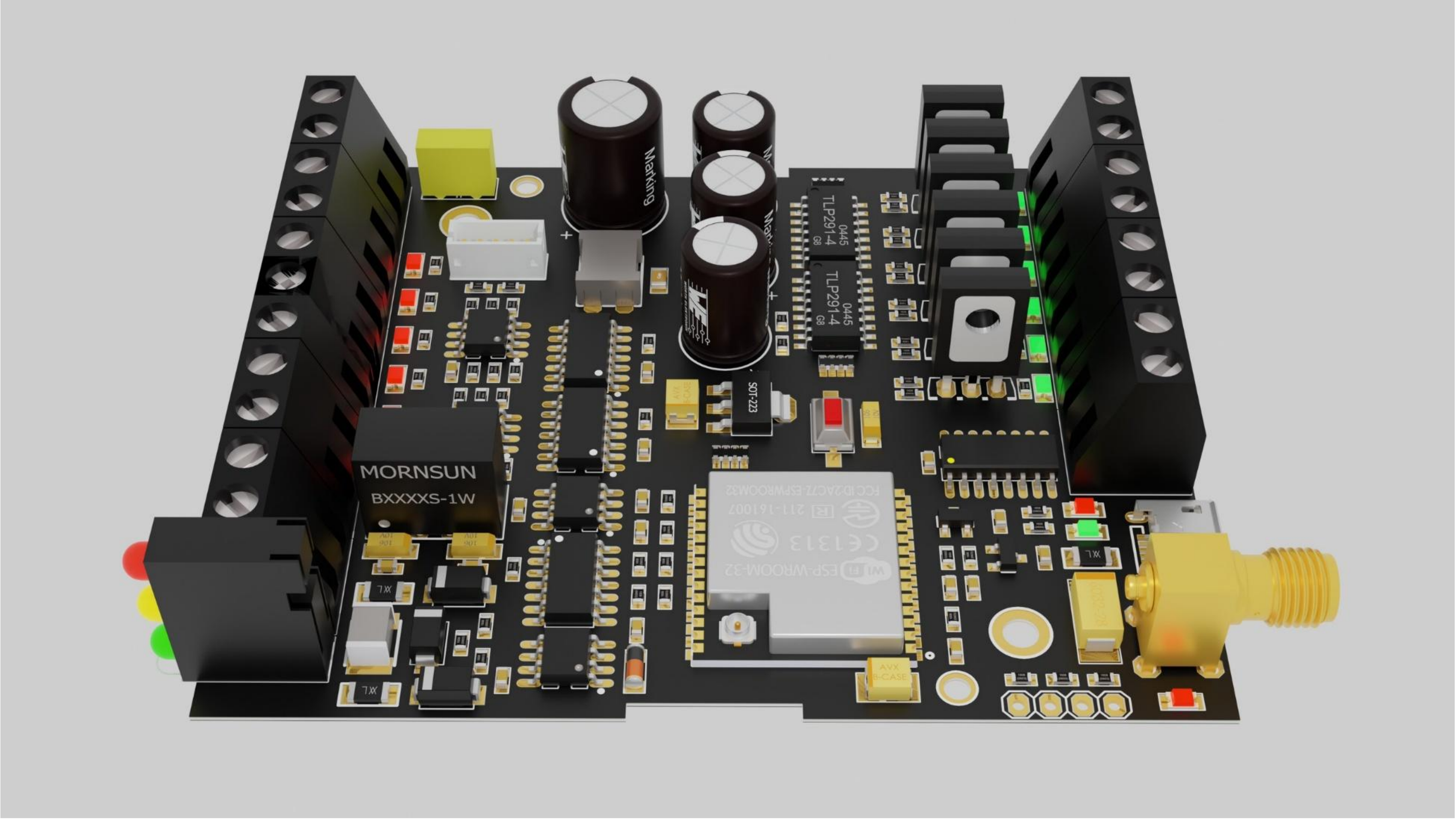
Chapter 4: Building an IoT System

Nguyen Tan No

04/2025

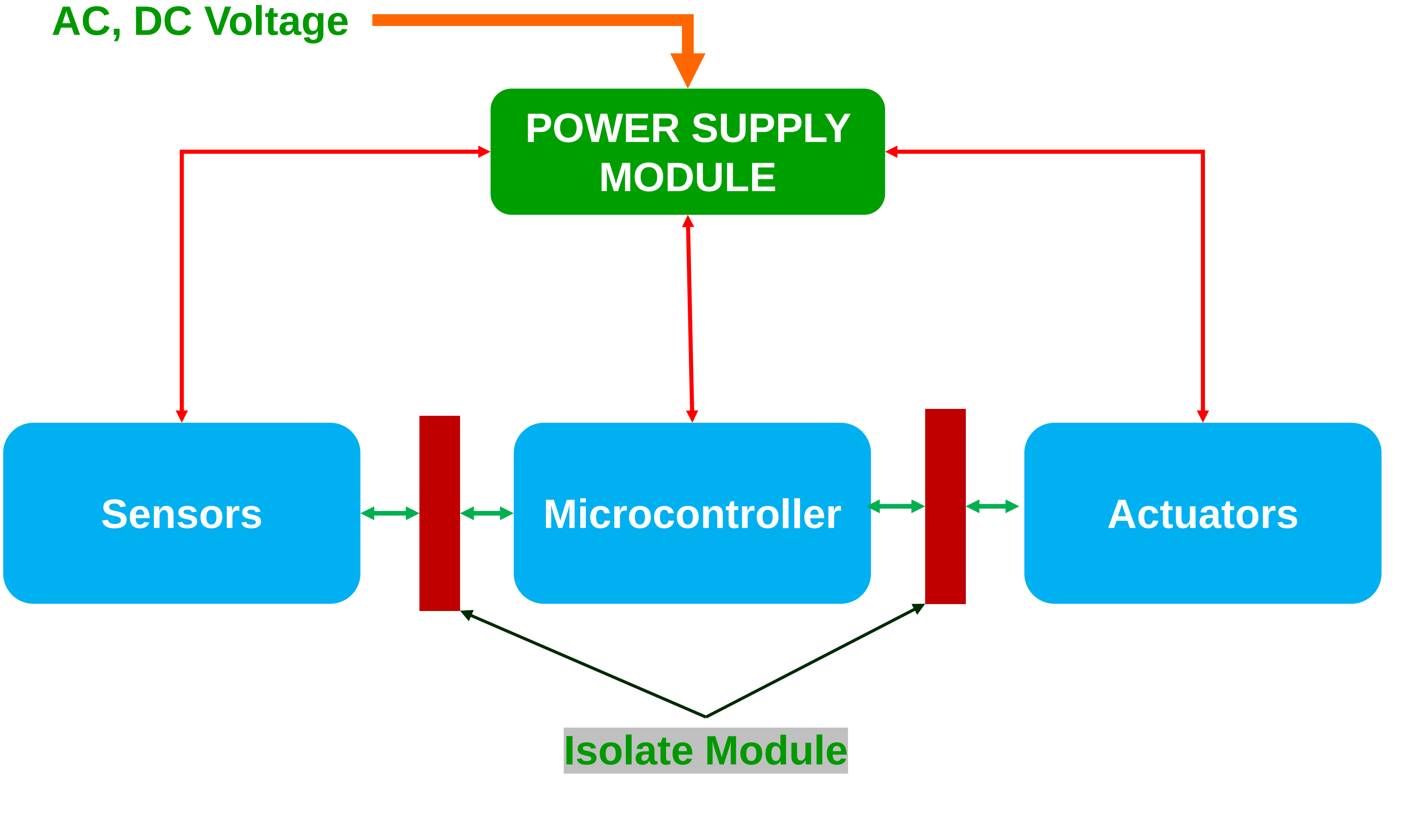
Contents

4.1. Hardware-Sensor Connection

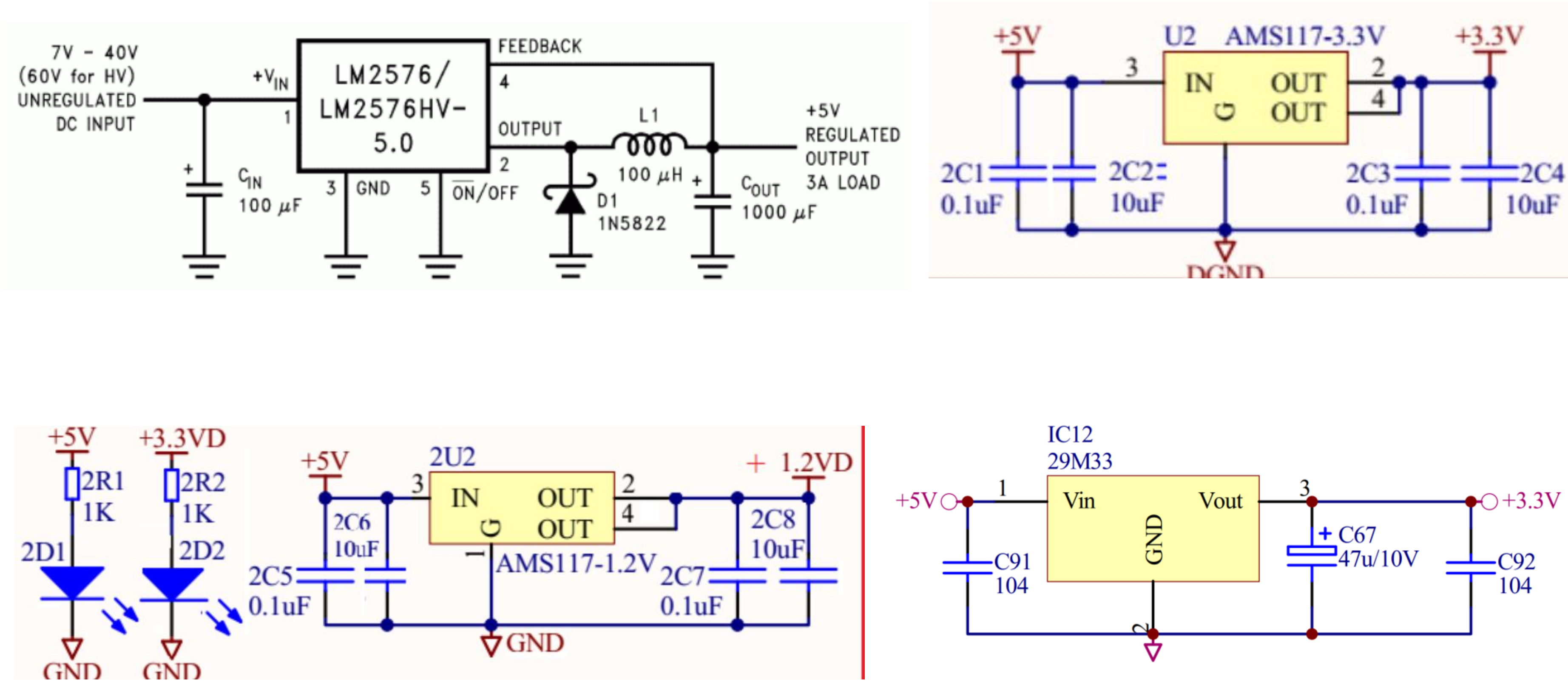


IoT Module

4.1. Hardware-Sensor Connection

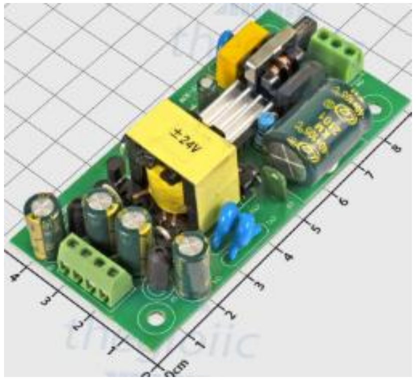


4.1. Hardware-Sensor Connection



Power Supply Shematic

4.1. Hardware-Sensor Connection



AC-DC

7805

LM2576

AMS1117

4.1. Hardware-Sensor Connection

Power Supply

sensor

Microcontroller

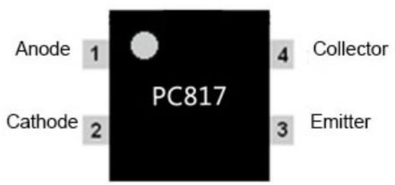
1.Digital

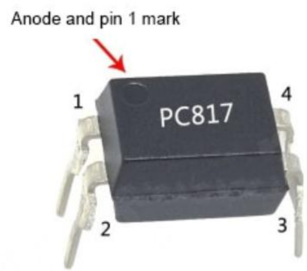
2.Analog

3.Communication

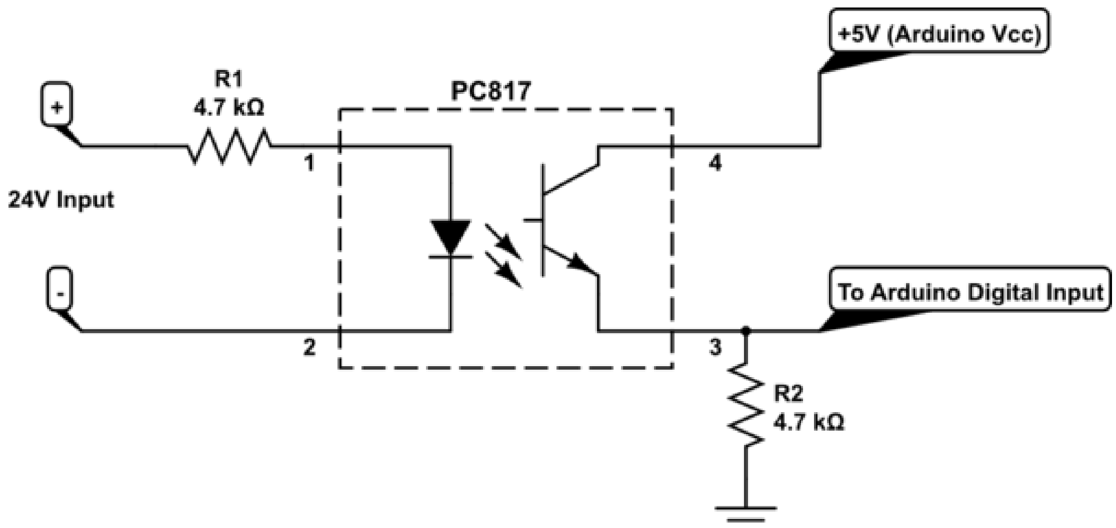
4.1. Hardware-Sensor Connection

1.Digital





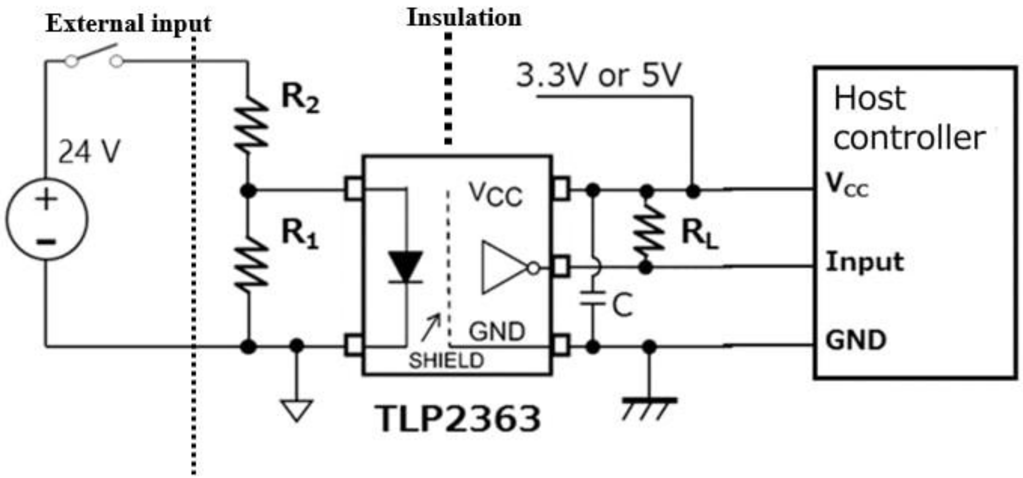
	Parameter		Symbol	Conditions
Input	Forward voltage		V_F	$I_F=20mA$
	Peak forward voltage		V_{FM}	$I_{FM}=0.5A$
	Reverse current		I_R	$V_R=4V$
	Terminal capacitance		C_t	$V=0$, $f=1kHz$
Output	Collector dark current		I_{CEO}	$V_{CE}=20V$
Transfer charac-teristics	*4Current transfer ratio		CTR	$I_F=5mA$, $V_{CE}=5V$
	Collector-emitter saturation voltage		$V_{CE(sat)}$	$I_F=20mA$, $I_C=1mA$
	Isolation resistance		R_{ISO}	DC500V, 40 to 60%RH
	Floating capacitance		C_f	$V=0$, $f=1MHz$
	Cut-off frequency		f_c	$V_{CE}=5V$, $I_C=2mA$, $R_L=100\Omega$, $-3dB$
	Response time	Rise time	t_r	$V_{CE}=2V$, $I_C=2mA$, $R_L=100\Omega$
		Fall time	t_f	

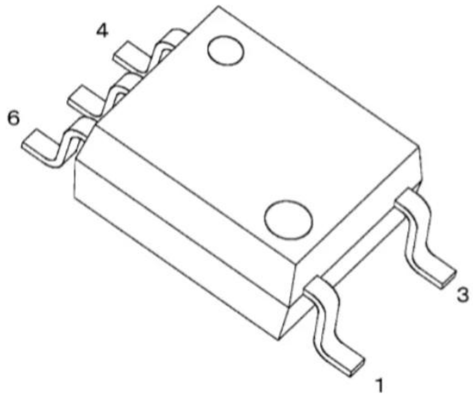


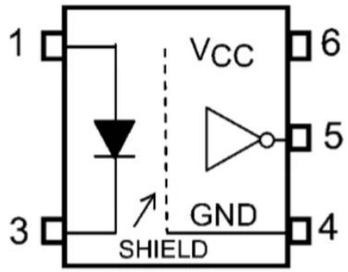
4.1. Hardware-Sensor Connection

1.Digital

- Inverter logic type (open collector output)
- SO6 packaging
- -40°C to +105°C operating temperature range
- 2.7V to 5.5V supply voltage
- 10Mbps (typ.) data transfer rate
- 4mA (max) supply current
- 0.3mA to 2.4mA (V_{CC} = 3.3V, R_L = 1kΩ) threshold
- ±20kV/μs (min) common-mode transient immunity
- 3750V_{RMS} (min) isolation voltage

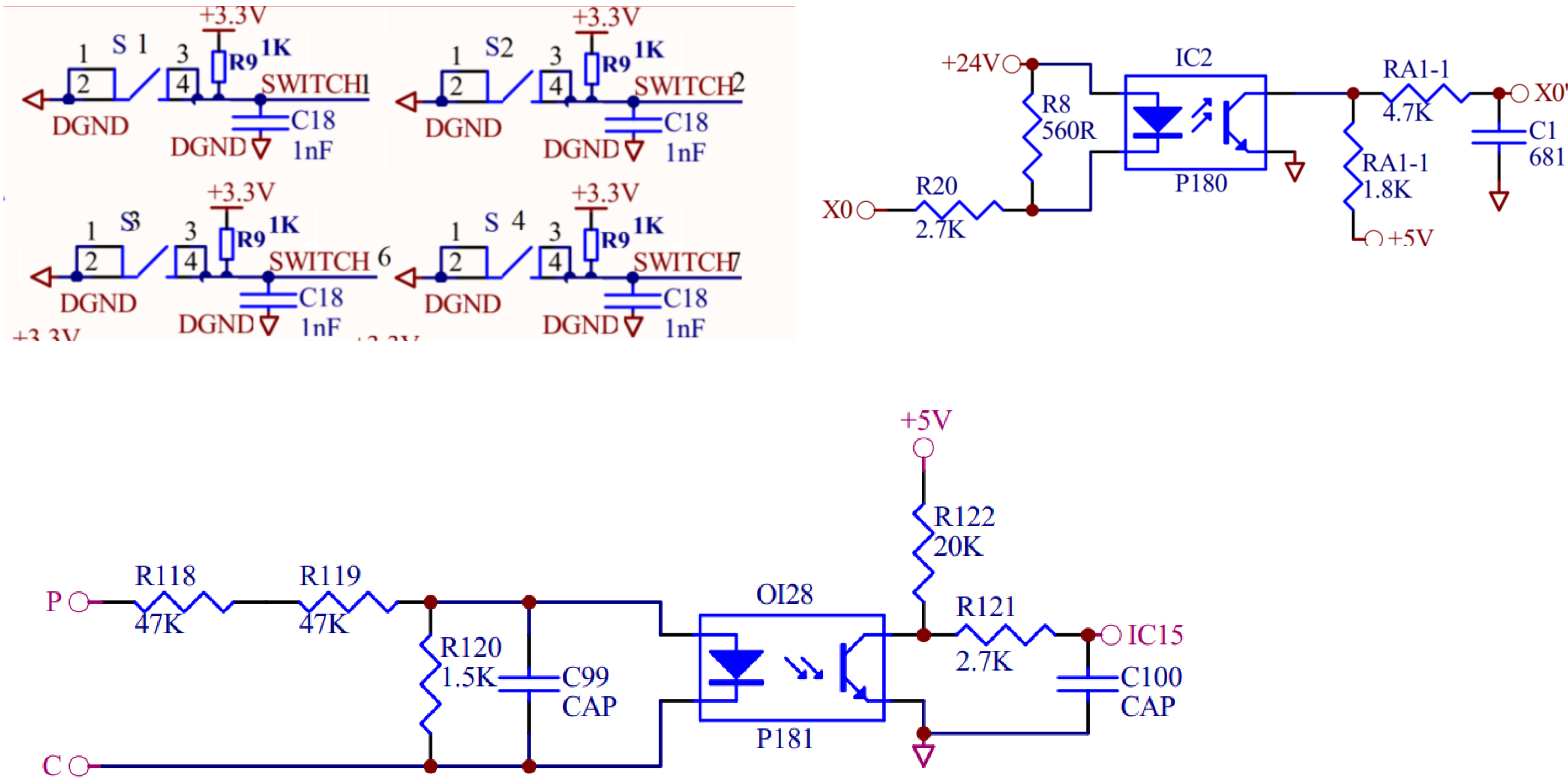






1: Anode
3: Cathode
4: GND
5: V_O (Output)
6: V_{CC}

4.1. Hardware-Sensor Connection



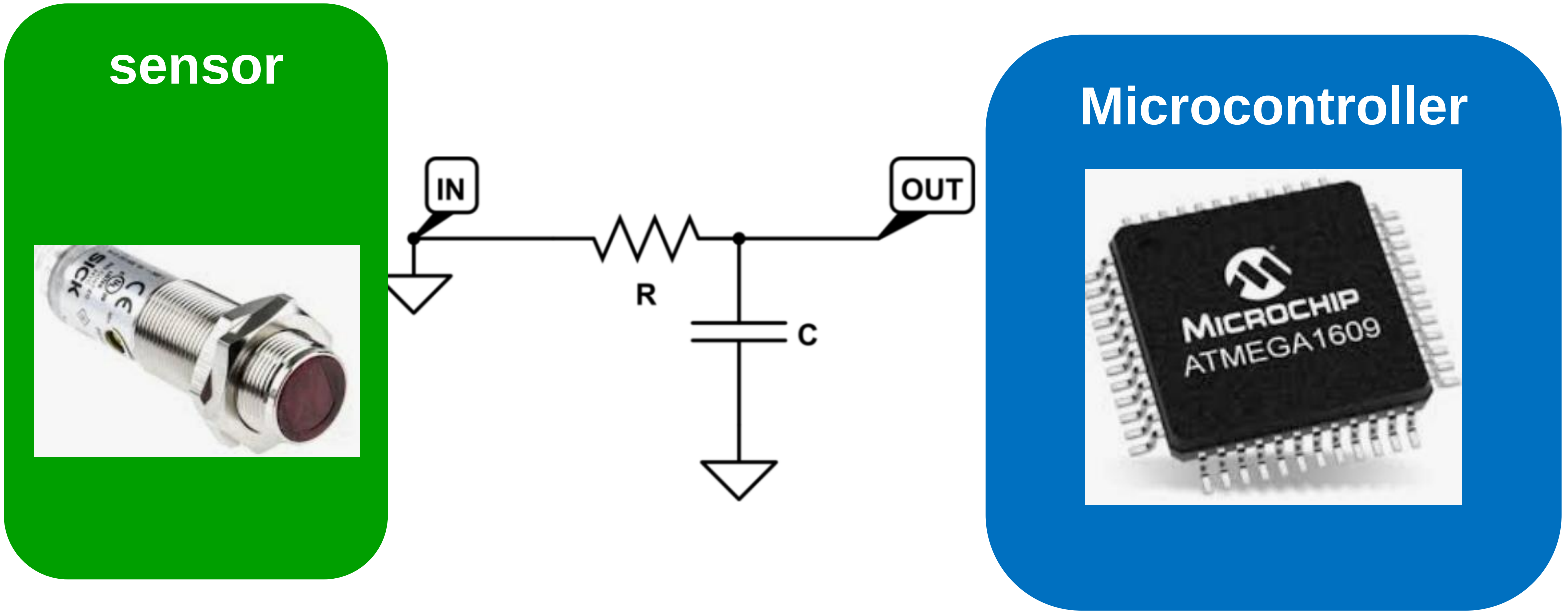
Input module shematics

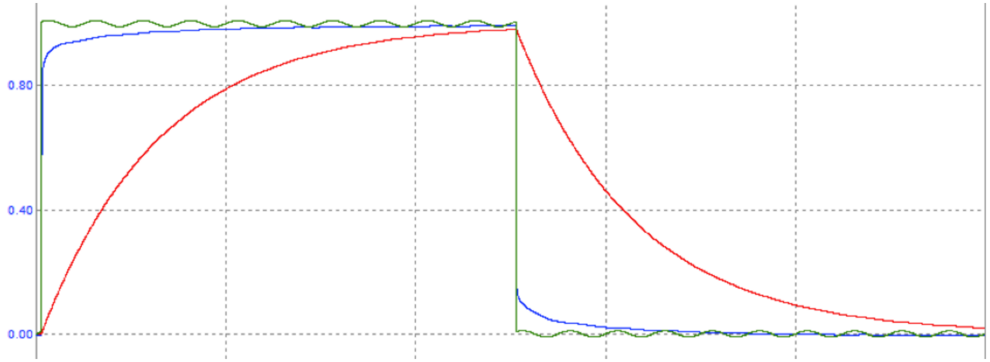
4.1. Hardware-Sensor Connection

Opto Module

4.1. Hardware-Sensor Connection

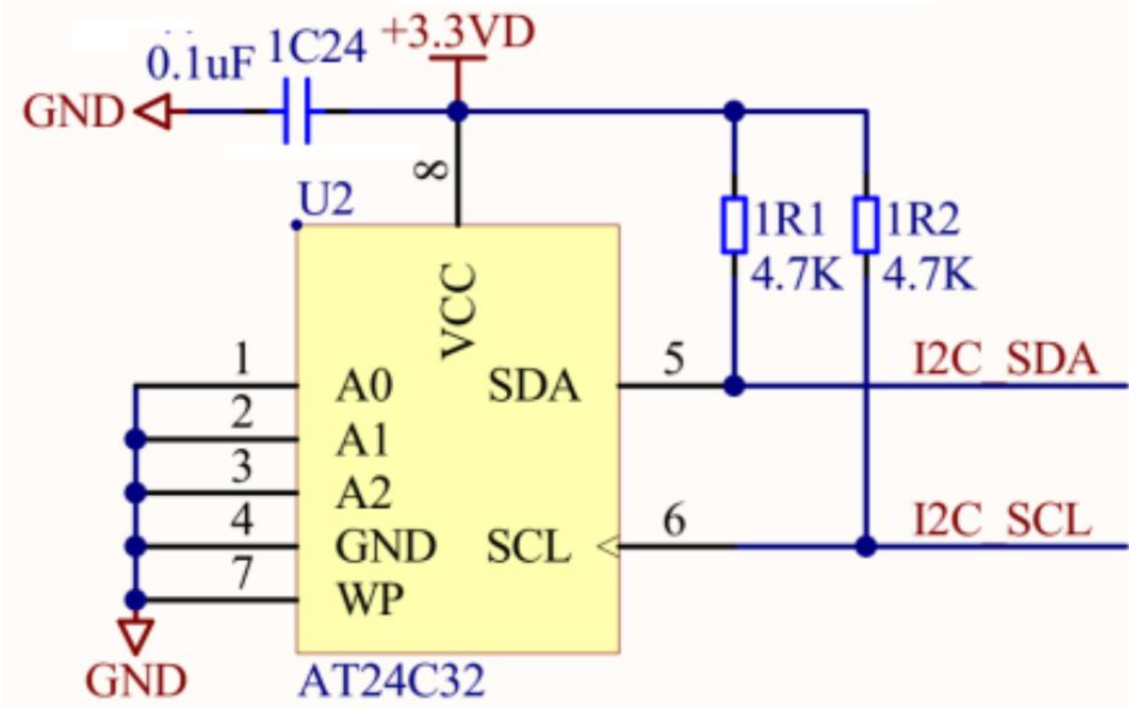
2.Analog





4.1. Hardware-Sensor Connection

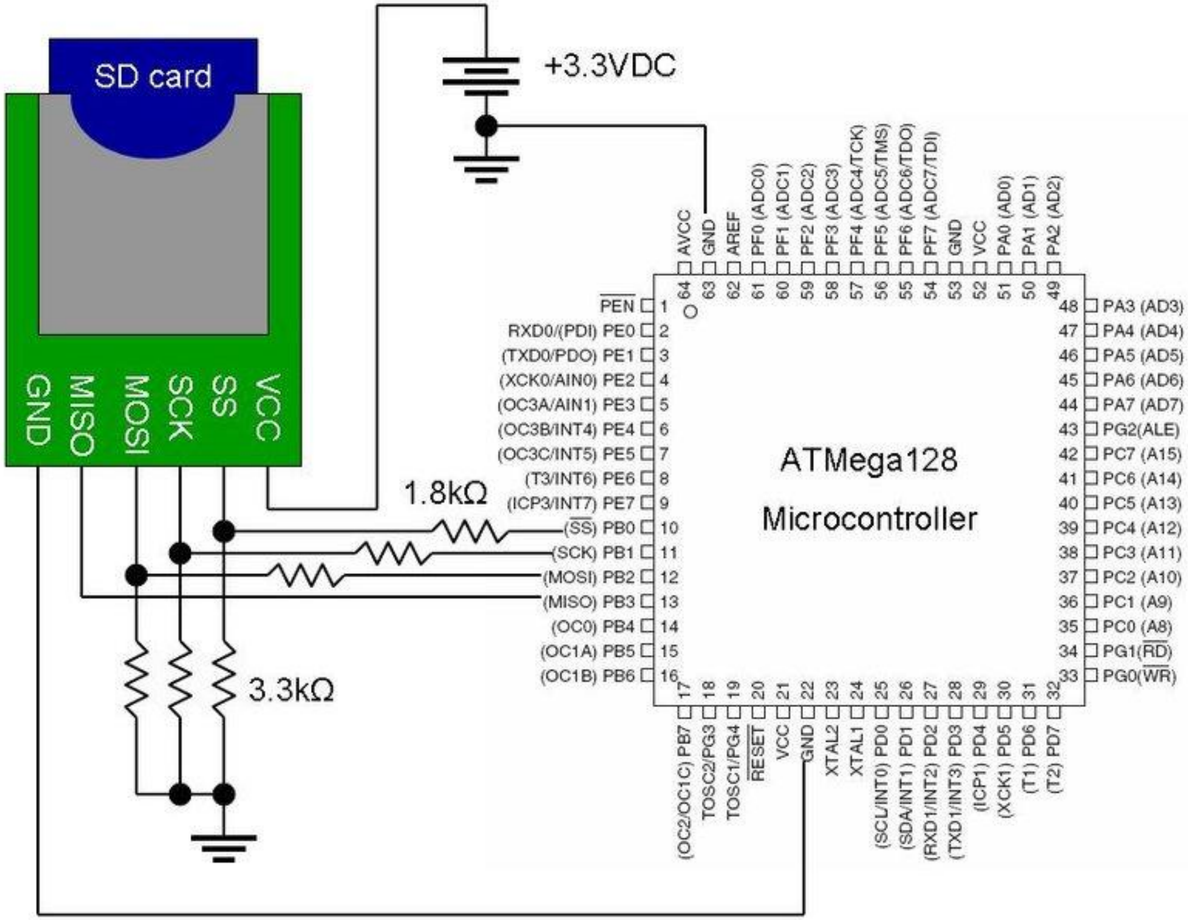
3.Communication



IIC

4.1. Hardware-Sensor Connection

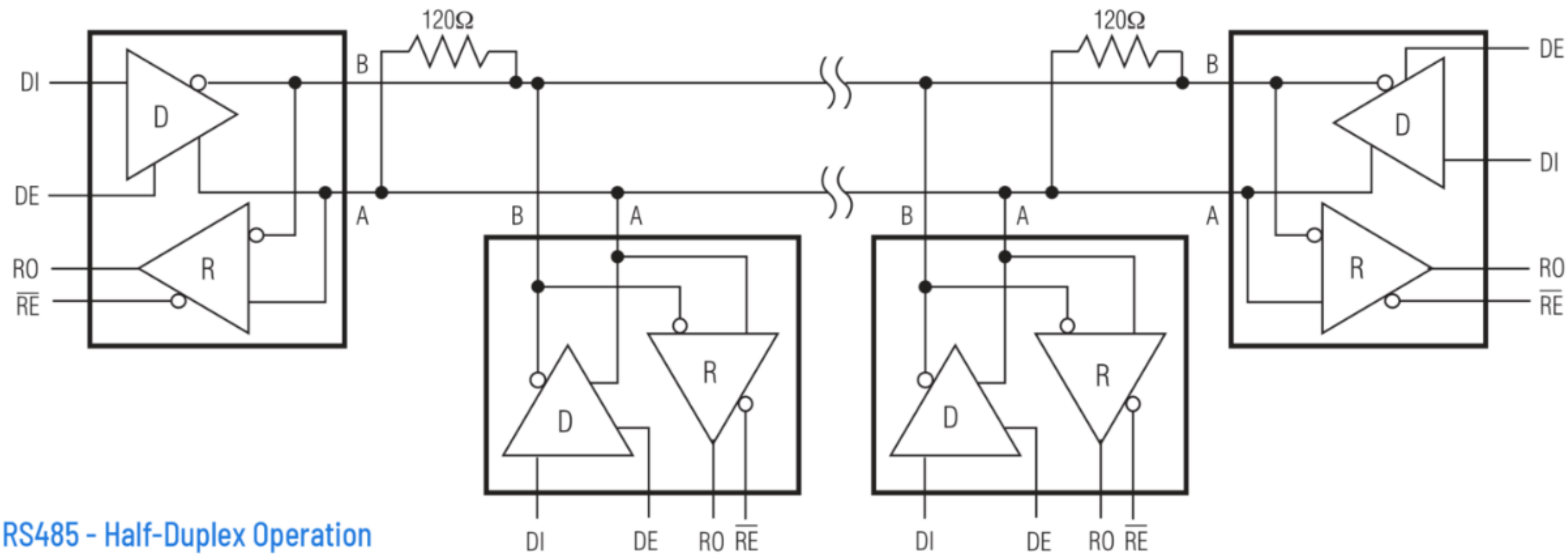
3.Communication

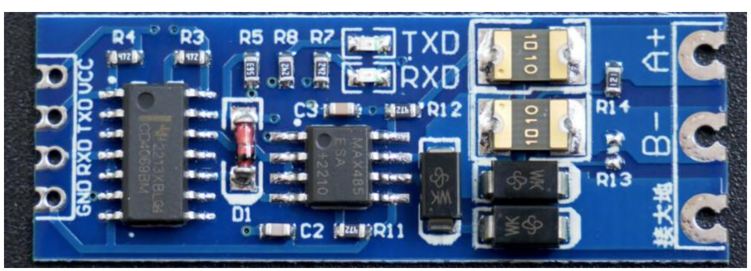


SPI

4.2. Hardware-Sensor Connection

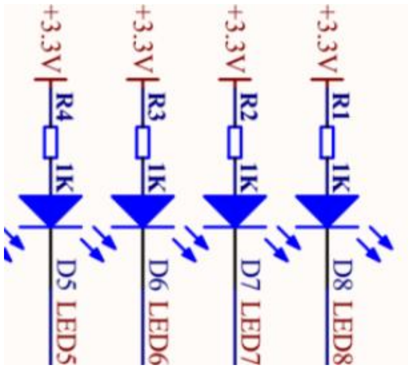
3.Communication



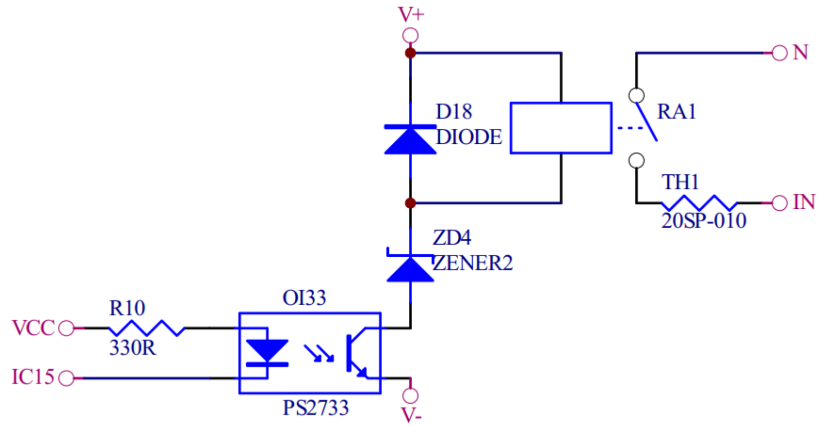


RS485

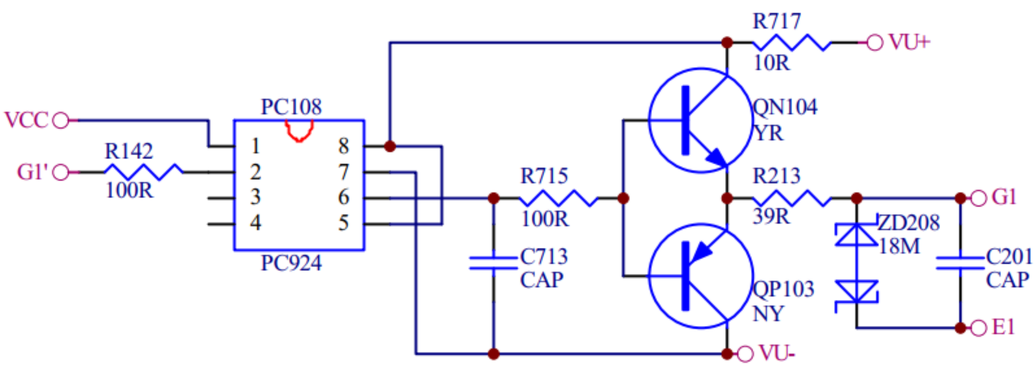
4.2. Hardware-Actuator Connection



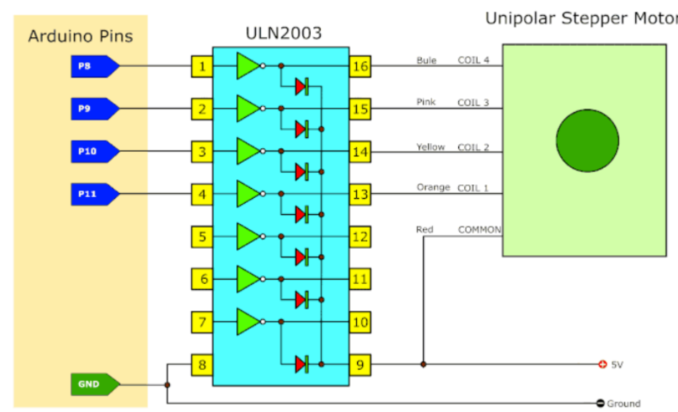
Led



Relay

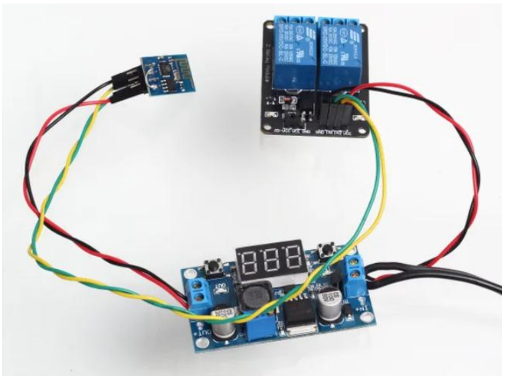


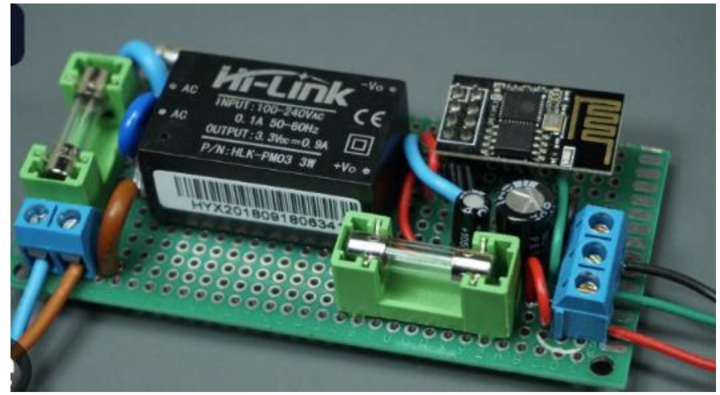
Transistor

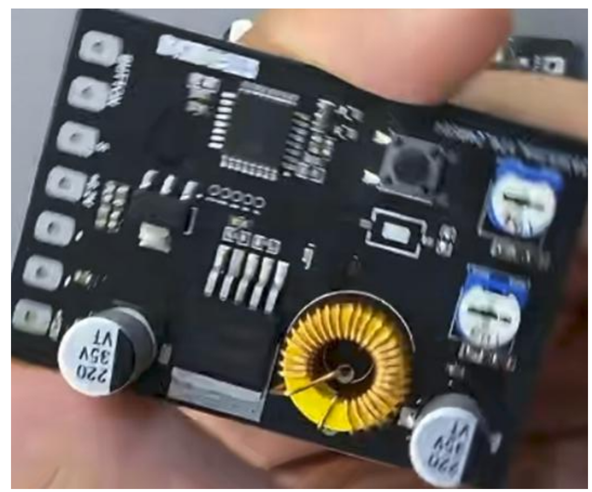


IC

4.2. Hardware-Actuator Connection

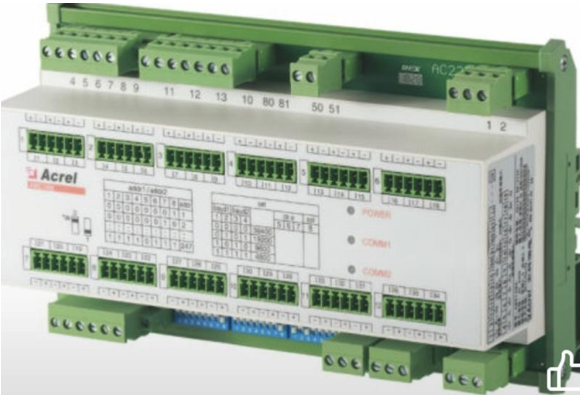






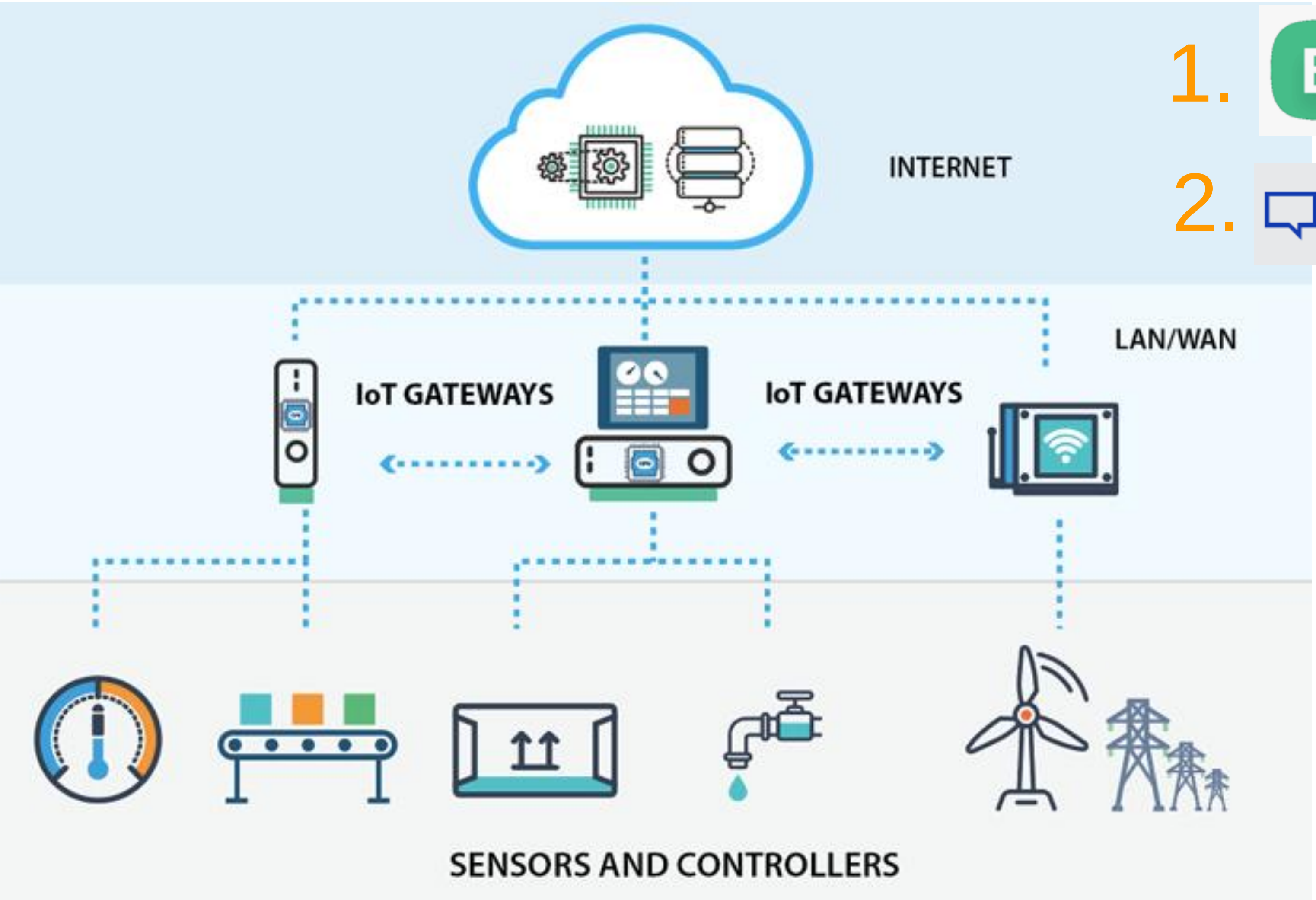
How to connect modules

4.2. Hardware-Actuator Connection



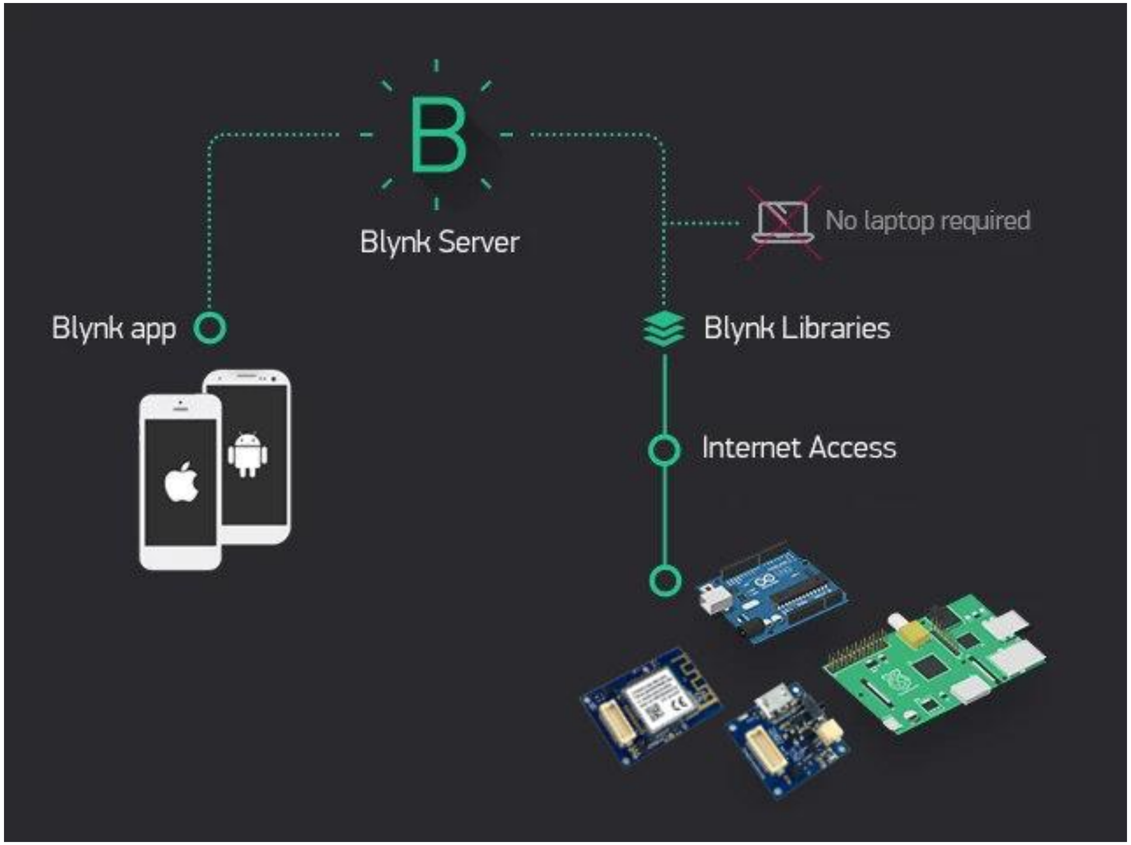
Product packaging

4.3. Programming cloud connectivity



1. Blynk
2. ThingSpeak

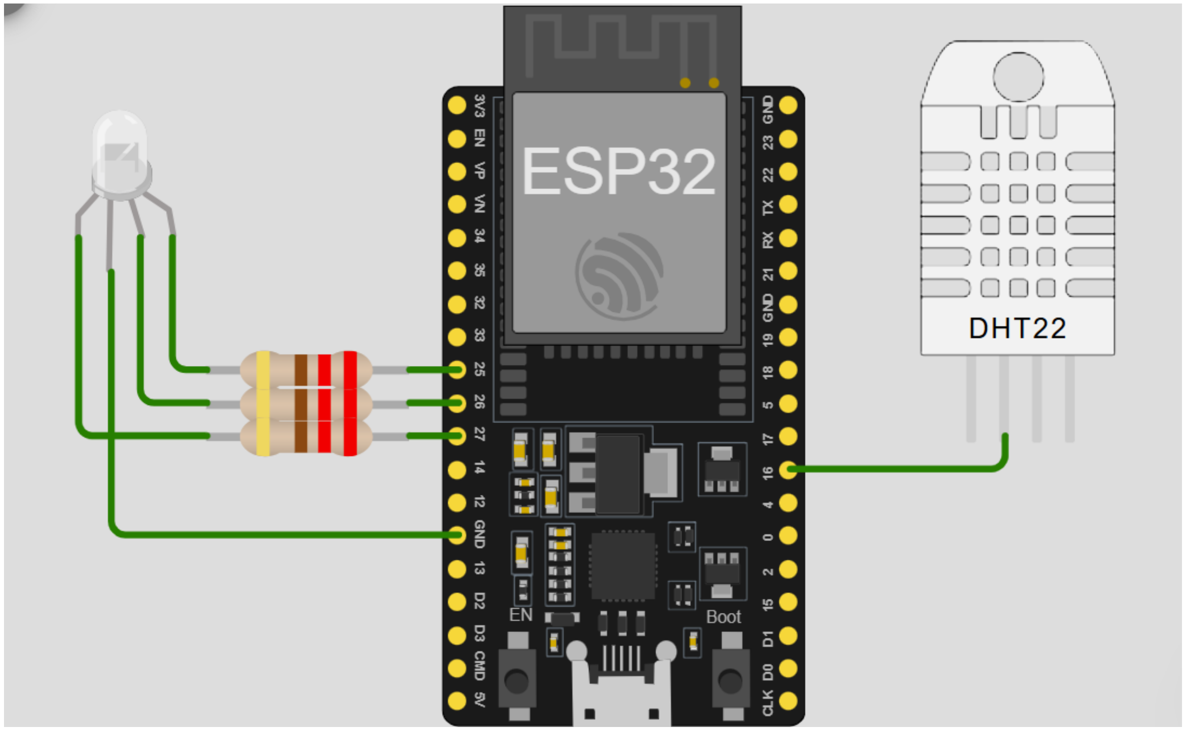
4.3. Programming cloud connectivity



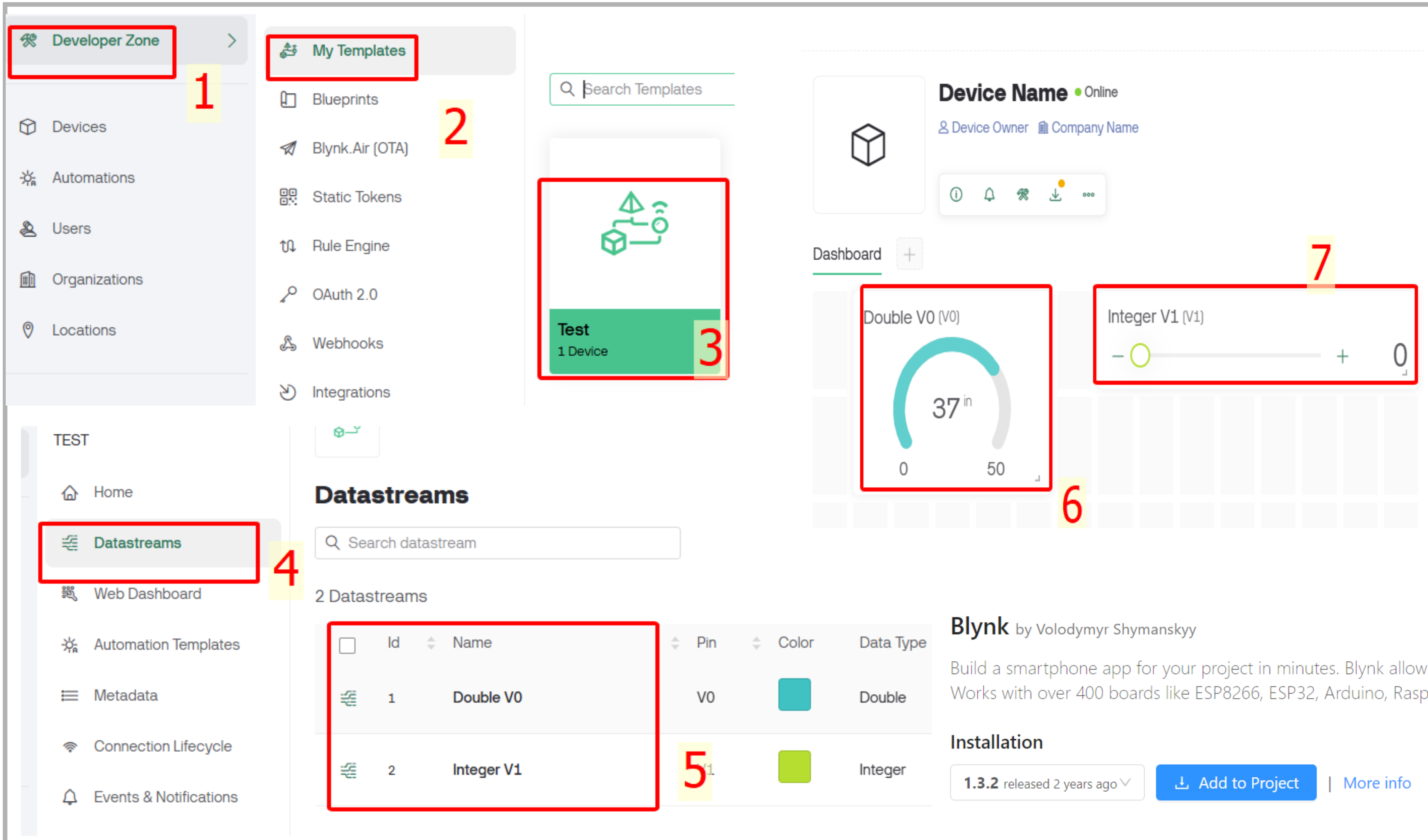
"Blynk is an IoT (Internet of Things) platform that helps users easily create remote control applications for smart devices. What makes Blynk special is its flexibility and ease of use."

4.3. Programming cloud connectivity

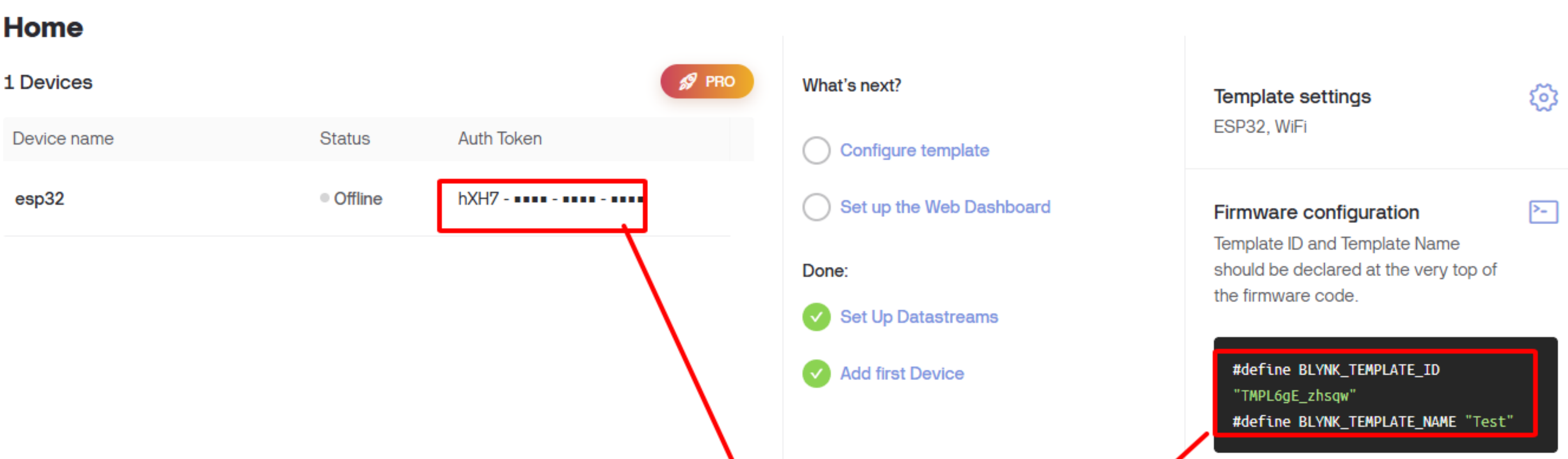
Project 1:



4.3. Programming cloud connectivity



4.3. Programming cloud connectivity




```
#define BLYNK_TEMPLATE_ID      "TMPL6gE_zhsqw"
#define BLYNK_TEMPLATE_NAME    "Test"
#define BLYNK_AUTH_TOKEN       "hXH7SulBBqnDCltIdjaKgiedn_H_-h7w"

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
char ssid[] = "Wokwi-GUEST";
char pass[] = "";
```

4.3. Programming cloud connectivity

```
void setup()
{
  Serial.begin(115200);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(1000L, myTimerEvent);
}
void loop()
{
  Blynk.run();
  timer.run();
}
```

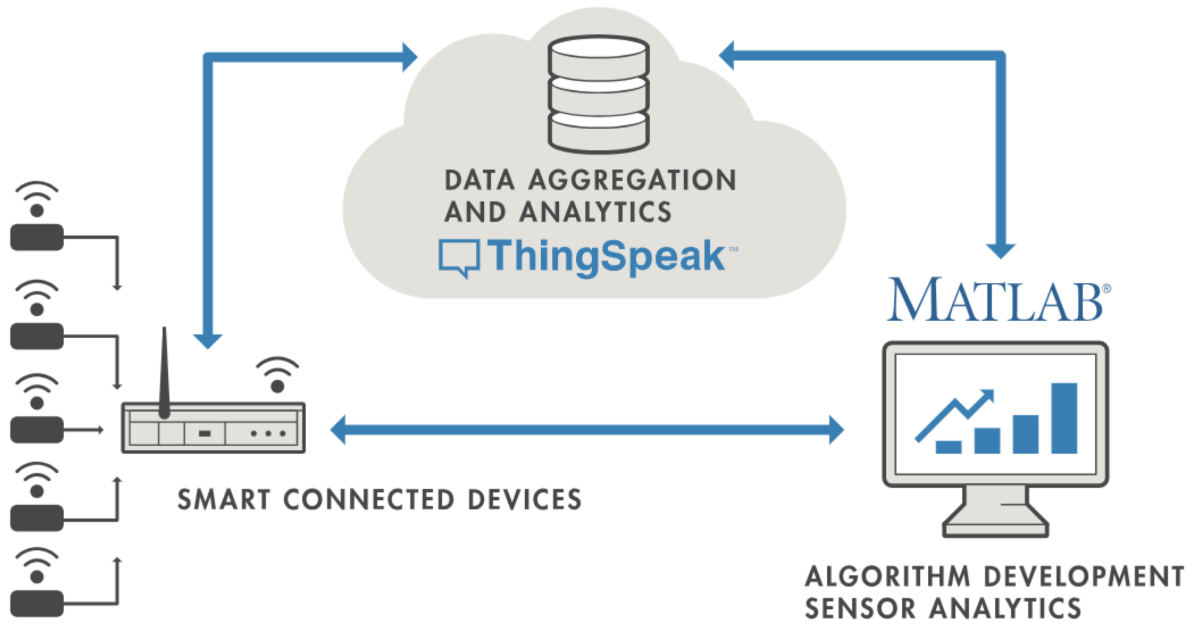
Read

```
BLYNK_WRITE(V1)
{
  int pinValue = param.asInt();
  Serial.println(pinValue);
}
```

Write

```
void myTimerEvent()
{
  counter+=3.4;
  Serial.println(counter);
  Blynk.virtualWrite(V0, counter);
}
```

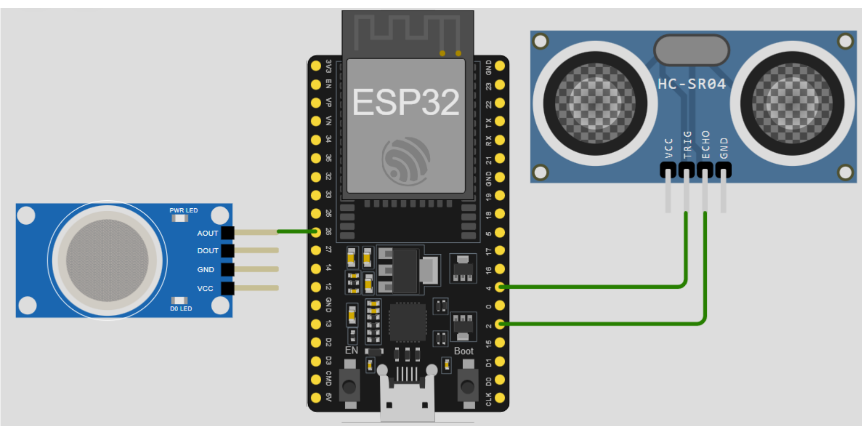
4.3. Programming cloud connectivity

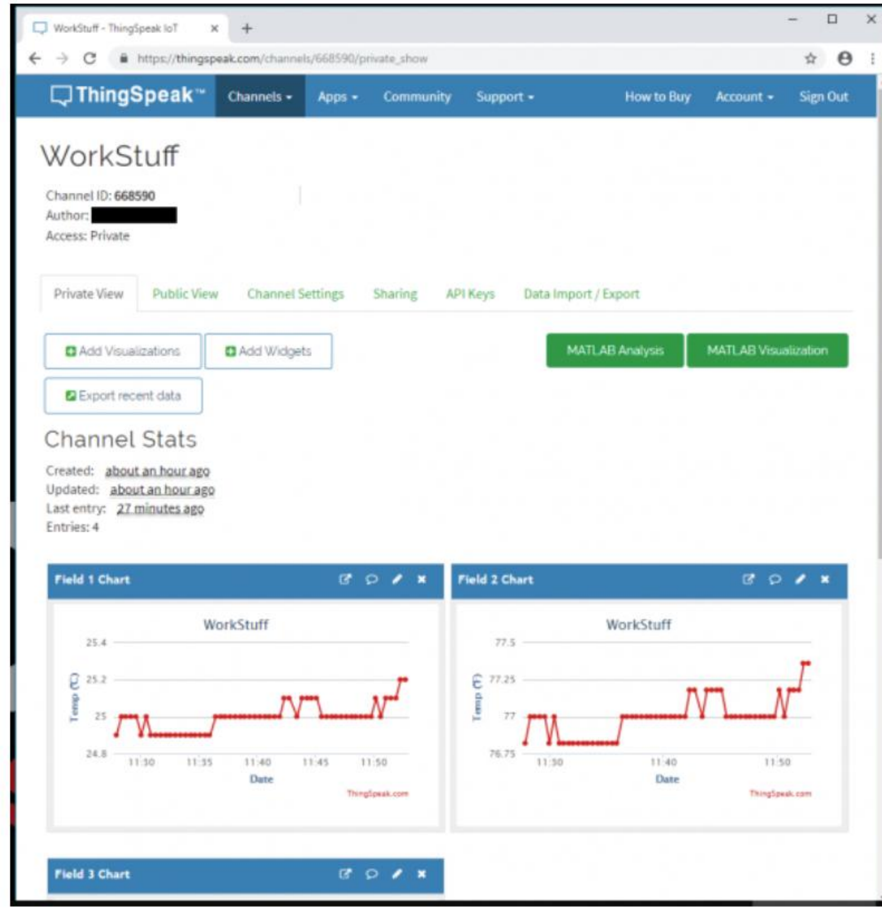


ThingSpeak™ is an IoT analytics platform service that allows you to aggregate, visualize and analyze live data streams in the cloud. ThingSpeak provides instant visualizations of data posted by your devices to ThingSpeak. With the ability to execute MATLAB® code in ThingSpeak you can perform online analysis and processing of the data as it comes in. ThingSpeak is often used for prototyping and proof of concept IoT systems that require analytics.

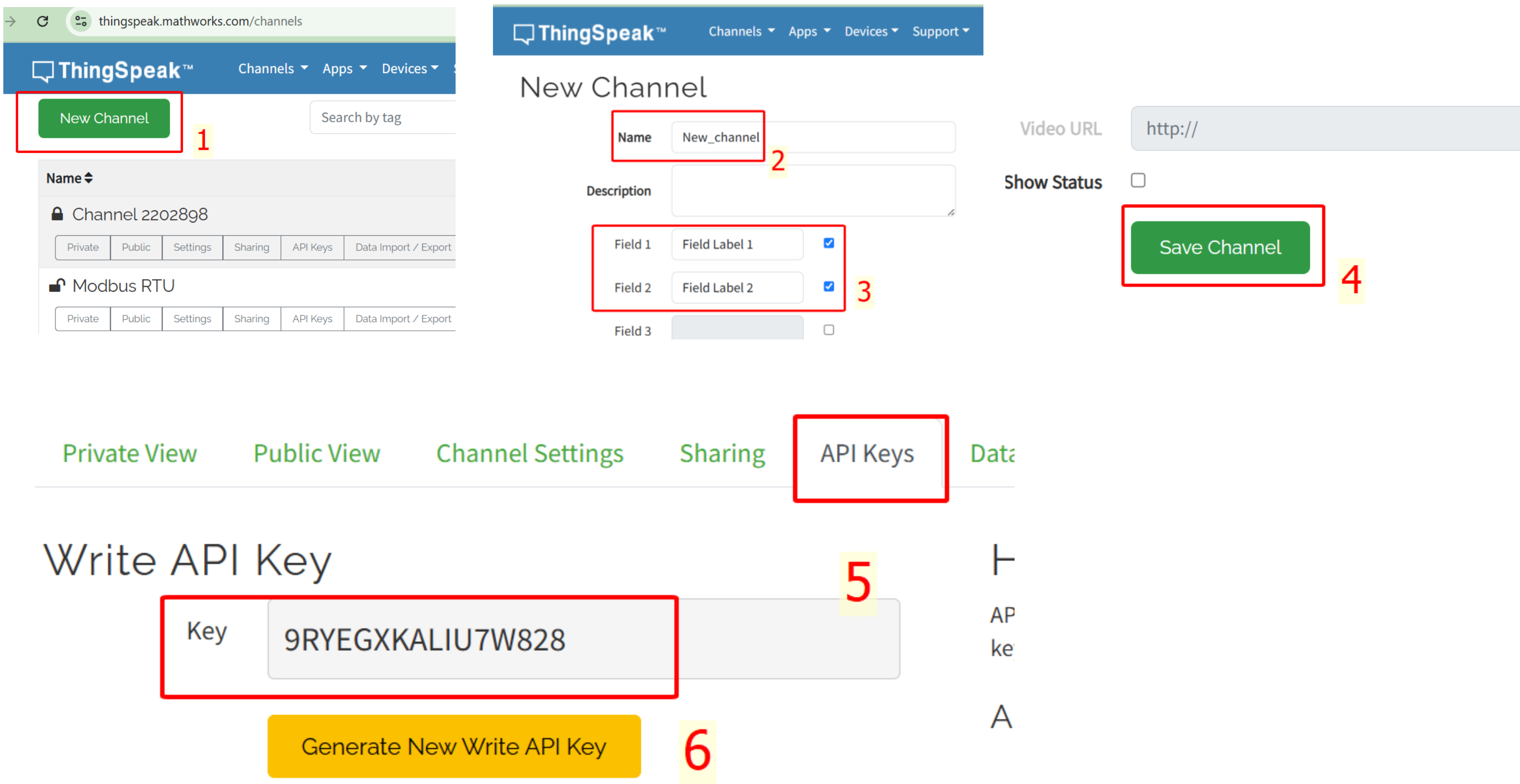
4.3. Programming cloud connectivity

Project 2:

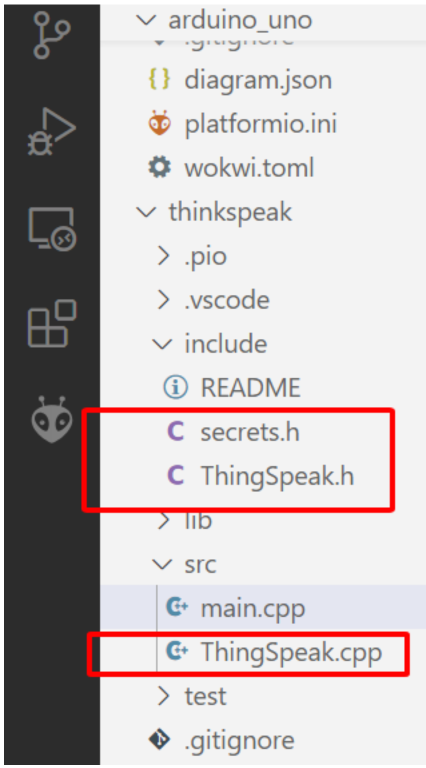




4.3. Programming cloud connectivity



4.3. Programming cloud connectivity

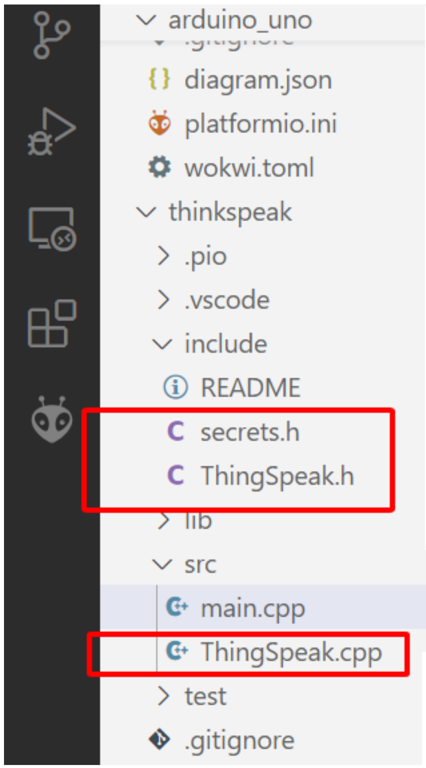



```
#include <WiFi.h>
#include "ThingSpeak.h"
char ssid[] = "Wokwi-GUEST";
char pass[] = "";

WiFiClient  client;
unsigned long myChannelNumber = 2923449;
const char * myWriteAPIKey = "9RYEGXKALIU7W828";

void setup()
{
  Serial.begin(115200);
  WiFi.mode(WIFI_STA);
  ThingSpeak.begin(client);
}
```

4.3. Programming cloud connectivity

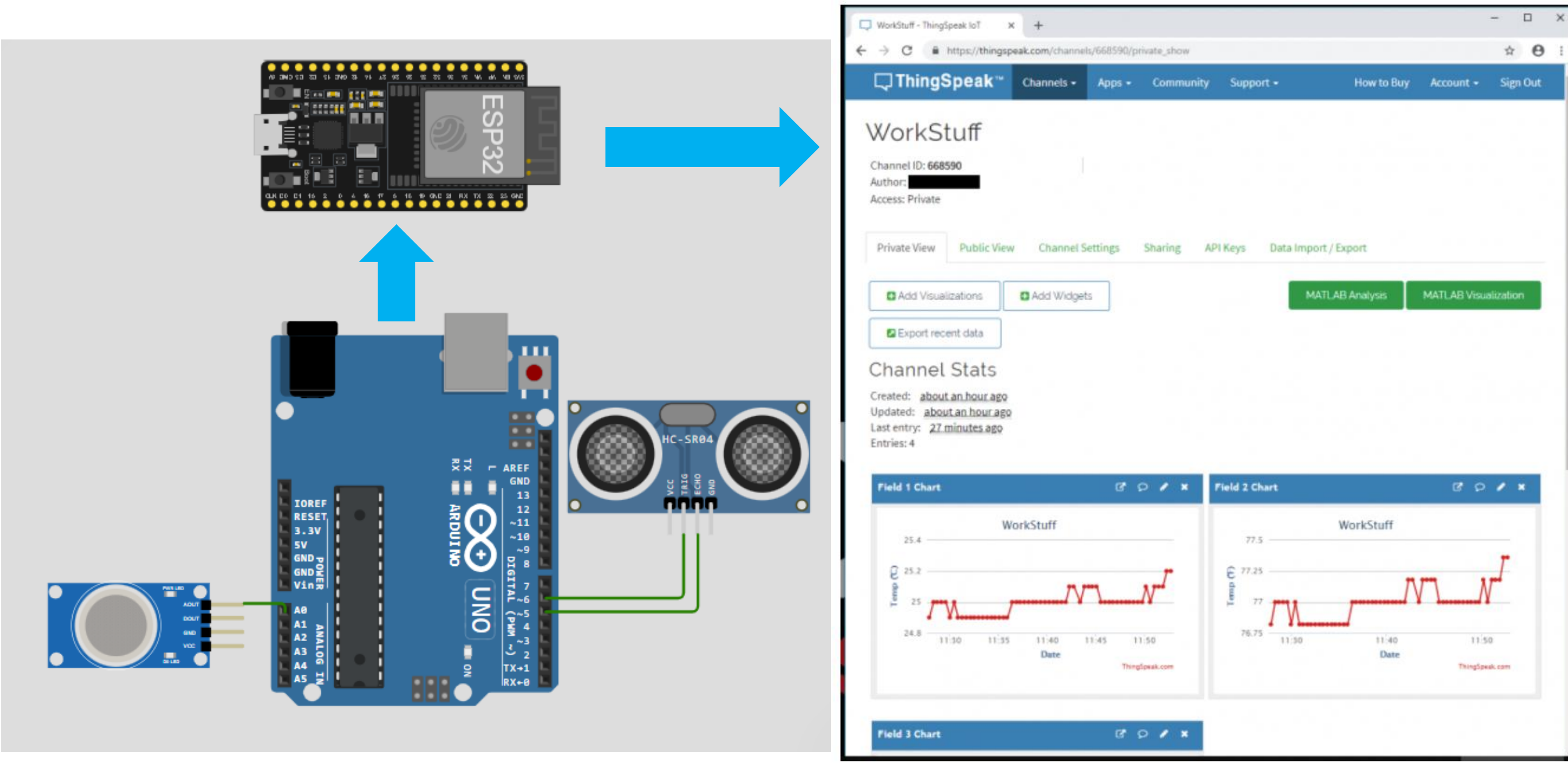



```cpp
// Connect or reconnect to WiFi
if(WiFi.status() != WL_CONNECTED)
{
  while(WiFi.status() != WL_CONNECTED)
  {
    WiFi.begin(ssid, pass);
    Serial.print(".");
    delay(5000);
  }
  Serial.println("Connected.");
}
```

```cpp
ThingSpeak.setField(1, number3);
ThingSpeak.setField(2, number4);
int x = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
if(x == 200)
  Serial.println("Channel update successful.");
else
  Serial.println("Problem updating channel. HTTP error code " + String(x));
```

4.3. Programming cloud connectivity

Project 3:



Thank for attention